

Birla Institute of Technology and Science, Pilani

Work Integrated Learning Programmes Division

M.Tech. in Artificial Intelligence and Machine Learning.

I Semester 2025-26

Course Number	AIMLZC416	
Course Name	Mathematical Foundations for Machine Learning	
Nature of Exam	Open Book	# Pages 3
Weightage for grading	30%	# Questions 7
Duration	150 minutes	
Date of Exam	//2025 (FN: 09:00 - 11:00)	

Instructions

1. All questions are compulsory
 2. Marks are indicated against each question
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- 1) Consider the following linear optimization problem

$$\begin{aligned} \min_{x_1, x_2} \quad & 3x_1 + 3x_2 \\ \text{subject to} \quad & 2x_1 + x_2 \leq 3 \\ & x_1 + x_2 \leq 1 \\ & 2x_1 + 3x_2 \leq 5 \end{aligned}$$

- (a) Rewrite this optimization problem in the following linear programming form learned in MFML course:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^2} \quad & \mathbf{c}^T \mathbf{x} \\ \text{subject to} \quad & \mathbf{Ax} \leq \mathbf{b} \end{aligned}$$

Mention clearly $\mathbf{A} \in \mathbb{R}^{3 \times 2}$, $\mathbf{c} \in \mathbb{R}^2$ and $\mathbf{b} \in \mathbb{R}^3$ in your solution.

- (b) Derive the Lagrangian of the reformulated optimization problem. Derive the minimum of Lagrangian with respect to $\mathbf{x}$.
- (c) Derive the dual function $D(\lambda_1, \lambda_2, \lambda_3)$ where $\lambda_1, \lambda_2, \lambda_3$ are the Lagrange multipliers.

- (d) Derive the dual optimization problem in terms of dual variables λ_1, λ_2 and λ_3 (1+2+2+1 marks)

Consider a binary classification problem where the two possible classes are +1 and -1. Assume that the linear classifier to be derived is of the form $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$. Here $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $b \in \mathbb{R}$. You are given the following three data points and their true classes $(\mathbf{x}_A, y_A)$, $(\mathbf{x}_B, y_B)$ and $(\mathbf{x}_C, y_C)$:

- 2) i) $\mathbf{x}_A = [1 \quad (4 + \sqrt{2})]^T$, $y_A = +1$
 ii) $\mathbf{x}_B = [(1 - \sqrt{2}) \quad 4]^T$, $y_B = -1$
 iii) $\mathbf{x}_C = [1 \quad 1]^T$, $y_C = -1$
- a) Write the SVM primal optimization problem for these data points by substituting the values of above $\mathbf{x}$ and y wherever needed. Ensure that all the inequalities have $\leq$ form.
- b) Derive the Lagrangian of the above optimization problem by introducing three Lagrange multipliers α_1, α_2 and α_3 . Simplify the Lagrangian by substituting the values of data points $\mathbf{x}$ and y wherever required.
- c) Write down the dual SVM optimization problem in variables α_1, α_2 and α_3 . Simplify the dual optimization problem by substituting the values of data points $\mathbf{x}$ and y wherever required.
- d) Derive the derivative of dual cost $Q(\alpha_1, \alpha_2, \alpha_3)$ with respect to α_1 . (2+1+2+1 marks)

Q3) Consider the following function

$$f(x) = e^{0.5x} + x^2 - 3x.$$

- (a) Verify whether the function $f(x)$ is convex on the interval $x \in [-2, 4]$.
- (b) Compute the critical points. Identify whether each critical point corresponds to a local minimum or maximum.
- (c) Apply the gradient descent algorithm to approach the minimum of $f(x)$. Take an initial guess $x_0 = 0.5$ and learning rate 0.1. Perform **five iterations** of the gradient descent method and record the successive values of x_k and $f(x_k)$. (2+2+2 Marks)

Q4) Consider the data given below.

5	3	6	2	8	7
2.5	2.2	3.0	1.8	3.4	3.1

- Standardize the two variables (rows) so that each has zero mean and unit variance. Display the standardized data matrix.
- Compute the covariance matrix of the standardized data and determine its eigenvalues and eigenvectors. Interpret the direction of the first principal component.
- Project the original (unstandardized) data onto the first principal component to obtain the one-dimensional representation of the six observations. (2+2+2 Marks)

Q5) We need to solve the system $Ax = 0$ where A is $m \times n$. It is known that $A = pq^T + rs^T$ for unit-length vectors p, q, r, s where $p \cdot r = 0$ and $q \cdot s = 0$. How many linearly independent solutions does the system $Ax = 0$ have? Show how to use Gram-Schmidt orthogonalization to solve this linear system. (6 Marks)

Q6) The Hessian of a function $f(x, y)$ turns out to be $\begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}$, no matter where the Hessian is computed. Assume that $(0, 0)$ is a critical point for f . Does there exist a unique $f(x, y)$ whose Hessian at any point is the given matrix? Give justification for your answer. For $\delta x, \delta y$ small enough to allow a second-order Taylor's approximation of f at the point $(0, 0)$, can you calculate the maximum difference $f(x + \delta x, y + \delta y) - f(x, y)$? (6 Marks)

Q7) a) If A and B are symmetric, prove or disprove that AB is also symmetric.

b) If A and B are symmetric and positive definite, prove using the following example that AB is not positive definite.

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 10 & 3 \\ 3 & 1 \end{bmatrix}$$

You may use a special vector $x = [0, 1]^T$ for showing the result.

c) If A and B are symmetric and $AB = BA$, then prove or disprove that the eigenspaces of A and B are the same. (1+2+1 Marks)